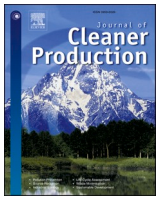
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From text to effectiveness: Quantifying green industrial policies in China

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ABSTRACT

The evolution of green industrial policy in China is deeply embedded within a unique political, economic, cultural, and social milieu. The intricacies and complexities inherent in this context present challenges for quantitative policy research. In the existing literature, the study of thematic clustering of green industry policy instruments and their effectiveness measures in large-scale policy text repositories remains a relatively unexplored area. To fill this research gap, this study provides an analysis of China's green industrial policies from 1994 to 2022. By adopting the Latent Dirichlet Allocation (LDA) topic model, implicit and optimal themes are identified, and key policy instruments are systematically and comprehensively parsed. Based on this, the cutting-edge Policy Modeling Consistency-Text Encoder (PMC-TE) index model was used to conduct an in-depth quantitative evaluation of various policy tools. The findings reveal 15 policy instruments from 7891 Chinese green industry policy documents. Notably, technical support, information resource demonstration project policies emerged as the most frequently deployed and efficacious. Conversely, government procurement and service outsourcing policies, while showing incremental growth year-on-year, lagged in utilization and effectiveness. These policy instruments can be broadly categorized into three primary archetypes: supply-type, demand-type, and environment-type. The supply-type instruments dominate, while environment-type instruments remain underutilized. Additionally, this study delineates the four chronological phases of China's green industrial policy evolution, illuminating China's strategic pivots-transitioning from resource allocation, to market dynamics, and ultimately to environmental governance trajectories.

1. Introduction

China is the world's largest energy consumer, driven primarily by its industrial sector and rapid urbanization (Naqvi et al., 2023). Concurrently, it remains the top global carbon emitter, largely due to its heavy reliance on coal for energy production (Tichenor et al., 2022; Yuan et al., 2020a, b, c). As the world grapples with the escalating challenges of climate change and sustainable development, the imperative to transition towards greener industrial practices becomes increasingly urgent (Karimi-Maleh et al., 2023a, b; Hayat et al., 2023a, b, c). There has been an unprecedented global focus on green industrial policies (Bo et al., 2023; Kyle, 2023). In global industrialization, China stands at a pivotal crossroads, where its industrial policies not only shape its economic trajectory but also exert a profound influence on the global environmental landscape (Qiu Yue et al., 2023). Given the escalating concerns around climate change and environmental challenges, transitioning from a tentative approach, the Chinese government has now fully embraced and expanded its commitment, leading to a multifaceted and mature green industrial policy framework spanning various sectors and levels (Steffen and Patt, 2022; Yuan et al., 2021). Policy creation and management have played a key role in the rapid development of China's green industry (Kiss and Popovics, 2021).

The one of the most crucial step in the creation and management of public policy is policy evaluation (Das et al., 2022). Policy evaluation is a multidimensional and comprehensive process that encompasses assessments of effectiveness, efficiency, impact, sustainability, and process (Hu et al., 2024; Ming, 2023). It focuses not only on the final outcomes of policies but also covers their entire life cycle, from design and implementation to their subsequent impacts, providing an all-encompassing analysis and reflection (Tiberghien, 2023). The value

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and significance of policy evaluation are manifold. Firstly, policy evaluation aids in understanding the motivations, design, and implementation specifics of policymaking, while also offering empirical support to policy theory (Chen et al., 2023a, b). It fosters innovation and diversity in research methodologies, including quantitative analysis, qualitative research, and mixed methods (Zhaohui and Yafei, 2022). Through an analysis of policy inputs and outputs, including a comprehensive assessment of different policy tools, regions, and stages, policy evaluation offers a methodological framework that makes policy research more precise and systematic (Usmani et al., 2023). Secondly, by evaluating the effectiveness and efficiency of policies, decision-makers can make more informed choices about adopting, modifying, or abolishing policies (Yang et al., 2023). Policymakers can adjust policy directions and execution strategies based on evaluation results, better adapting to societal needs and environmental changes (Wang and Zhang, 2023). Moreover, by identifying the most effective policy tools and practices, it ensures that limited resources are allocated to areas that yield the most positive impact (Yeo and Oh, 2023). Especially in complex social environments with multiple stakeholders, systematic policy evaluation helps ensure the achievement of policy objectives.

China's green industrial policy is a complex system, involving politics, economy, science and technology, culture, society, technology and other fields (Wei et al., 2020). Although the intention of policy formulation is often clear, extracting and quantifying the effectiveness of policy implementation from a large number of policy texts faces multiple challenges (Ghasem et al., 2023). So far, most of the previous literature on the evaluation of China's green industry policies evaluates policy effects from the macro level, paying little attention to the quantitative analysis of a single policy tool, and lacking classification research on China's green industry policy tools. Particularly where there may be interactions and overlaps between various policy instruments, it is difficult to distinguish and measure the degree of quantification of each instrument. Moreover, most of the existing literature uses qualitative evaluation methods, and their conclusions are often highly subjective, contain personal biases, and lack objectivity and scientificity. As the complexity of policy formulation and implementation increases, how to ensure that the research methodology is adaptive and forward-looking to keep pace with changing policy environments and needs is a key issue.

To address these prominent challenges, the innovations and contributions of this study are underscored in the following four aspects: First, leveraging policy encoding strategies in tandem with cutting-edge machine learning methodologies, this research has crafted a comprehensive, long-spanning database of China's green industrial policies, which is vast in quantity and broad in coverage. Second, utilizing Latent Dirichlet Allocation (LDA) topic modeling and natural language processing, this research have been able to accurately identify and extract the core elements, key terms, and instruments within these policies. This methodology has enabled us to cluster policy tools effectively, minimizing the risk of overlooking or duplicating tools in analysis. Third, this research has developed and applied a novel framework for assessing policy effectiveness, integrating the Policy Modeling Consistency-Text Encoder (PMC-TE) index model. This approach expands upon the traditional Policy Modeling Consistency (PMC) index model, providing a more nuanced evaluation that considers various influencing factors. This enables a more precise measurement of each policy tool's effectiveness, enhancing the rigor and depth of the findings. This study also delves into the historical development and evolutionary trends of China's green industrial policy tools. Through this exploration, this research provided new insights into the progression and patterns of policy evolution, offering valuable perspectives for both researchers in policy studies and policymakers.

This study is organized as follows. Section 2 reviews the literature. The methods and data are described in the third section3. Section 4 provides the results. Section 5 presents a discussions. Section 6 summarizes conclusions and future directions.

2. Review of literature

2.1. Research on green industrial policy

Green industrial policy is centered on resolving and coordinating the contradiction between regional development and the environment (Giordano, 2015). It is a series of related industrial policies that tend to support industrial low-carbon, emission reduction and green transformation and development (Fan et al., 2020; Liu and Liu, 2022). Green industrial policies uniquely intersect at the crossroads of industrial development and environmental sustainability (Zhengguang et al., 2023a, b). The genesis of green industrial policy in China dates back to the 1990s (Huaping et al., 2019). These policies are crafted not merely to support industrial activities indiscriminately, but to specifically prioritize and incentivize industries contributing to environmental goals such as reduced carbon emissions, sustainable resource utilization, and improved energy efficiency (Philip et al., 2023). They often encompass a mix of regulatory measures, fiscal incentives, and support for research and innovation in green technologies (Dan et al., 2022). Additionally, these policies might set explicit targets, such as reducing greenhouse gas emissions in manufacturing, increasing the share of renewable energy in industrial production, or encouraging traditional industries to adopt clean technologies (Mandy, 2021; Yuan et al., 2020a, b, c). Within the Chinese context, the ambit of green industrial policy includes initiatives specifically aimed at steering the industrial sector towards sustainable development (Xin et al., 2023). This includes policies supporting renewable energy sectors like solar and wind power, implementing stricter environmental regulations in traditional industries, and encouraging the adoption of green technologies and practices in manufacturing and production processes (Matthias van den and David, 2023; Karaman et al., 2022; Yuan et al., 2022).

Green industrial policies are grounded in a solid theoretical basis for promoting green growth, and despite facing practical implementation challenges, these hurdles are not insurmountable (Chang and Andreoni, 2020). Extensive research indicates that judicious green industrial policies can drive the optimization of industrial structures, foster technological innovation, and facilitate a low-carbon economic transition, thereby achieving dual benefits for the economy and the environment (Ghosh et al., 2022; Yuan et al., 2021). Furthermore, studies by researchers like Philip and colleagues on green industrial policies in European countries demonstrate that policy-driven incentives can channel private investment into green industries, accelerating the development of a green economy (Xin et al., 2023). However, there is also a spectrum of scholarly opinion that expresses reservations about the efficacy of green industrial policies (Mølstad and Karseth, 2016). Researchers like Demidov et al. have pointed out that in some countries, an over-reliance on government subsidies for green industries can lead to industrial bubbles and misallocation of resources (Zhaotian et al., 2023; Victor., 2023). It has been found that in countries lacking a robust market mechanism and legal framework, green industrial policies may struggle to achieve their intended effects (Aikens et al., 2018).

In summary, green industrial policies represent a dynamic and evolving field, replete with both opportunities and challenges. While they offer a promising path towards sustainable industrialization and environmental preservation, their success is contingent upon balanced policy design, effective implementation, and a supportive market and legal environment.

2.2. Research on evaluation of green industrial policy

The body of literature focusing on evaluation of China's green industrial policies typically falls into two distinct categories. The first is a qualitative evaluation of green industry policies, and the second type of literature focuses on the quantitative evaluation of the implementation effects of individual green industry policies. The first type of literature is relatively abundant, and most scholars use case law to sort out policy

content to identify differences and synergies between different policies (Hamilton, 2017; Di Giovanni and Rogers, 2023). Zahedi, R. focused his research on case studies of Iran's green energy policies and strategies. By reviewing policy documents and official reports, he assessed the potential of renewable energy sources (Zahedi et al., 2022). Some scholars use content analysis to count the frequency of specific words or phrases in policy texts to capture the key themes or issues of the policy (Dandan and Delu, 2022; Duy et al., 2023; Cao et al., 2022). Tang K. and colleagues conducted a macro-level strategic analysis of factors influencing corporate green innovations, delving into the positive and negative ramifications of green industrial policies, and highlighting the underlying reasons and challenges of policy implementation (Gao and Sebastiani, 2016; Tang et al., 2020). The second type of literature mainly explores the effectiveness or effect of individual green industrial policies through econometric methods. Matheus K. and his team measured the effects of renewable energy policies in Latin American and Caribbean nations using economic indicators like grid pricing, loans, and taxation (Matheus et al., 2022). Dongyang Z. and colleagues adopted a key planning directive issued by the Chinese government as a temporal variable milestone, using the total factor productivity of enterprises as the explanatory variable to verify the effectiveness of green industrial policies for energy companies(Dongyang and Qunxi, 2022). Wu, S. and his team applied the Difference-in-Differences (DID) approach, incorporating indicators like economic growth, external financing, capital stock, and labor, to quantitatively evaluate the outcomes of green credit policies based on panel data (Wu et al., 2022).

2.3. Literature gaps

Firstly, most of the existing policy analyses had predominantly relied on traditional qualitative analytical methods or econometric methods. Such methods had often fallen short of delving deep into and interpreting the ambiguous, complex, and multilayered phrasing within policy texts. Moreover, few studies had embarked from policy texts themselves to categorize policies through clustering analysis. Instead, the prevailing categorization approaches had been largely empirical, with a tendency to directly adopt policy tool classifications from other domains, such as technological innovation.

Secondly, China's green industry policies, having been nurtured and evolved within specific political, economic, cultural, and social contexts, had posed challenges for quantification due to the subtle shifts in these backgrounds and the intrinsic complexities. The current literature on policy efficacy assessment had mainly hinged on expert scoring methods. The inherent subjectivity of this approach had possibly compromised the objectivity and fairness of the results. Furthermore, when substituting a particular dimension of a policy with a specific economic indicator for evaluation, the selection variance among different researchers had led to considerable uncertainty or heterogeneity in the outcomes.

To address the gaps in existing research, this study employs an integrated machine learning approach combining LDA modeling, policy coding, quantitative tabulation, and PMC-TE modeling. This approach enables a cluster analysis of policies, a thorough exploration and comparison of various policy tools, and vividly reveals the evolutionary patterns of green industrial policies under different categories. This study not only expands the methodological approaches in policy analysis theoretically but also provides policymakers with new tools and perspectives that have practical guidance for policy formulation and evaluation. From a theoretical standpoint, this research breaks through the limitations of traditional qualitative analysis and econometric methods by integrating advanced machine learning technologies like the LDA and PMC-TE models. This methodological innovation not only enhances the precision of policy analysis but also enriches the theoretical framework of policy research. Traditional studies often rely on subjective judgments to interpret policy texts. In contrast, this study utilizes advanced text mining technologies to interpret complex and multi-layered information within policy texts more accurately. From a practical perspective, the quantitative analysis methods employed in this research reduce the subjectivity inherent in expert scoring methods, thereby increasing the transparency and objectivity of policy evaluation. This makes the assessment of policy effects more scientific and credible. Furthermore, the classification and comparison of policy tools can assist policymakers in better understanding the strengths and limitations of various instruments, leading to the formulation of more effective policies.

3. Methodology and data

In this study, this research introduced a framework leveraging LDA and PMC-TE to quantitatively assess the efficacy of China's green industrial policies while elucidating the trends in policy instrument evolution. First, this research gathered pertinent policy documents from the designated domain, followed by a meticulous cleaning and organization process rooted in the inherent context of the policy texts. After that , utilizing Natural Language Processing (NLP) and LDA topic model, this research accurately identified the core essence and optimal topics within the policy texts, extracting various categories of policy instruments. Moreover, this research designed a quantitative guideline and innovatively adopted the PMC-TE index model. This model not only quantifies policy instruments but also precisely captures the intricate interactions between influencing factors. Finally, this research effectively differentiated and measured the specific effects of each policy instrument, revealing the evolution of the policies, thereby ensuring the precision and depth of the research conclusions. The methodology framework of the study is shown in Fig. 1.

3.1. Data

Based on the provided definition of green industrial policy, this study uses industry, emission reduction, low carbon, energy saving, dual carbon and green as search terms for this policy text until September 16, 2023, through the Beida Legal Dictionary System and the China Law Search System. This research obtained 8163 policy documents, including 521 central laws and regulations, 7511 local laws and regulations, 9 legislative materials, 9 Chinese and foreign contracts, 1 Taiwan law and regulation, and 110 legal developments.

Policy text culling was then carried out, eliminating invalid and less relevant policy texts, while excluding policy documents with only key words and no clear initiatives (Tichenor et al., 2022). If the same policy appeared in more than one year, the policy content was compared, and if the policy content was consistent or updated only in terms of year, the one with an earlier year was retained. Because this study needs to quantitatively study the policy texts of complete years, 2023 is less than one year, so the policy texts of 2023 are deleted, and the collection of policy texts is up to December 31, 2022. In the end, this research got 7891 Chinese green industrial policies (Table 1).

3.2. Identifying and clustering of policy instruments

Latent Dirichlet Allocation (LDA) topic model is a statistical model that applies algorithms to semantic analysis and clustering of corpus to discover text topics and learn them unsupervised (Liu et al., 2022). LDA topic model was proposed by Blei et al. Currently, LDA-based topic model methods have been applied in the fields of natural language processing, text mining, policy classification, and information retrieval, etc (Jelodar et al., 2019). Unsupervised learning LDA is a typical three-layer Bayesian bag-of-words model, including document layer, topic layer, and topic word layer (Blei et al., 2003). Each document and each topic corresponds to a keyword probability distribution.

This research provided an important basis for understanding the underlying structure and intent of the policy-making by constructing a LDA topic model to identify optimal policy topics and extract different

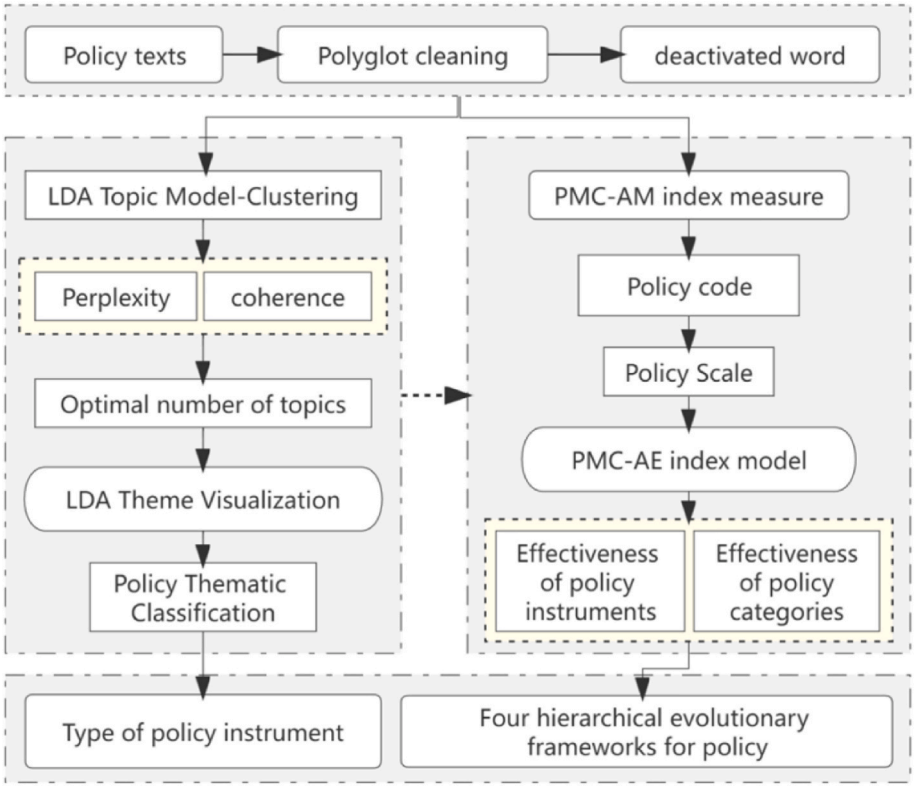


Fig. 1. Methodology framework.

Table 1
Key messages of selected Chinese green industry policy texts.

Number	Policy name	Regulatory Authority	Issue Number	Date of Issue	Active Rank	Timeliness
1	Reply of the State Council on Agreeing to the Construction of National Sustainable Development Agenda Innovation Demonstration Zone in Huzhou City	State Council (PRC)	State Letter [2022] No. 70	2022.07.10	State Council Normative Documents	currently in force
2	Regulation on China (Anhui) Pilot Free Trade Zone	Anhui Provincial People's Congress (including Standing Committee)	Anhui People's Congress Standing Committee Bulletin No. 61	2022.03.25	Provincial local regulations	currently in force
3	The CPC Central Committee and the State Council issued the Outline of the Plan for the Construction of the Chengdu-Chongqing Twin-city Economic Circle	Central Committee of the Communist Party of China	not have	2021.10.20	The Party's Internal Regulations System	currently in force
4	Statistical Classification of Energy Saving and Environmentally Clean Industries (2021)	International Bureau of Statistics	National Bureau of Statistics Decree No. 34	2021.07.26	departmental regulation	currently in force
5	Notice of the Office of the People's Government of Shuozhou City on the Issuance of the Organization Plan of Huashuo Green Industry Development Group Co.	Shuozhou Municipal People's Government	Shuozhengbanfa [2020] No. 51	2020.11.18	Local normative documents	currently in force
6	Shanxi Banking and Insurance Supervision Bureau's Circular on Further Strengthening Green Finance Work	Shanxi Banking and Insurance Supervision Bureau	Jin Yin Insurance Supervision and Administration [2020] No. 49	2020.07.28	Local normative documents	lose effectiveness
7	Approval of the National Development and Reform Commission on the Issuance of Corporate Bonds by Fujian Fuzhou Urban Construction Investment Group Co.	PRC National Development and Reform Commission (NDRC), formed in 2003	Development and Reform Enterprise Bond [2019] No. 30	2019.03.20	Administrative License Approval	currently in force
8	Circular on the Overall Program for the Construction of a Financial Open Portal Facing ASEAN in the Gangxi Zhuang Autonomous Region	People's Bank of China	Yinfa [2018] No. 345	2018.03.15	Sectoral working papers	currently in force
…	…	…	…	…	…	…
7891	Circular of the General Office of the Hebei Provincial Government on the Issuance of the Outline of Agricultural and Rural Economic Development in Hebei Province in the 1990s	Hebei Provincial People's Government	Ji Zhengban [1994] No. 17	1994.09.22	Local working papers	currently in force

kinds of policy instruments. LDA is particularly suited for handling large volumes of text data, effectively distinguishing and identifying different policy instrument topics. Unlike some traditional methods, LDA does not require predefined topics or categories. It can discover topics automatically from the data, making the research more objective and comprehensive. LDA can handle texts of various lengths and formats, making it well-suited for diverse data sources such as policy documents, reports, etc. In this research, LDA can reveal previously unnoticed policy themes and patterns, providing new perspectives for policy analysis. This research can conduct more effective comparative studies across different policies and periods.

The process of LDA mainly includes literature partition cleaning, determining the optimal number of topics and extracting thematic terms, topic degradation and visualization analysis. The prior distribution Dirt(β) is used by the LDA topic model to construct the topic-word distribution φ_k of K topics. The topic sequence of the documents is then determined by randomly generating the document-topic distribution θ_m of M documents from another prior distribution Dirt(α). The word order of the documents is then determined by adding the number of words D_m of a document derived from the Poisson distribution and the topic-word distribution φ_k.The word order of the documents is then determined by adding the number of words of a document derived from the Poisson distribution and the topic-word distribution. The cleaned data is analyzed by LDA topic model algorithm to determine the optimal number of topics. The document-topic distribution and topic-word distribution are measured at the same time, the clustering results are visualized, and the topic classification is realized by combining the topic distribution and word semantics.

In the LDA topic model, it is assumed that the document is composed of multiple topics and each topic is in turn composed of multiple words. First, the value of the number of topics k in the model needs to be specified by domain analysis and cross-validation (Chen, 2023, Chen et al., 2023b). The two hyperparameters α and β of the LDA topic model were made to be set to 50/k and 0.01 (Ruiz Estrada et al., 2007; Harrison et al., 2017). Then, the optimal number of topics is determined by evaluating the topic model through the performance metrics Perplexity and Coherence, which are commonly used in LDA topic model.

Consistency is used to evaluate the quality of the topics generated by the LDA topic model. Consistency takes into account the semantic similarity between the most relevant words in a topic. The number of high-frequency topics in the text is extracted from the topic consistency score, and the higher the consistency score, the better the topic quality. Perplexity is a metric that evaluates the ability of a topic model to predict new data. Mathematically, for a test set, perplexity can be viewed as the geometric mean of the inverse probability of the model. A smaller perplexity means that the model clusters better. The confusion curve is obtained through formula (1).

$$\text{Perlexity}(D_{test}) = \exp\left\{ -\frac{\sum_{i=1}^{M} Inp(d_i)}{\sum_{n=1}^{M} N_i} \right\} \quad (1)$$

where M is the number of documents and N_i is the number of words in document i, and $p(d_i)$ is the likelihood of document i.

In this study, based on the LDA topic model, this research evaluated the filtered policy texts for advanced perplexity and consistency to determine the optimal number of topics for green industrial policies. Then an unsupervised machine learning mining strategy was initiated for data extraction, which in turn led to a fine-grained visualization of the LDA topic model of China's green industrial policies. By identifying theme feature words, this research categorize multiple policy instruments under green industrial policy, which are further decomposed into different strategy categories.

3.3. Quantitative guide for policy instruments

This research designed a measurement model for green industrial policy, as shown in Table 2. The green industrial policy measurement model systematically deconstructs this multi-objective and complex policy problem, splits it into multiple dimensions, and then quantifies it based on specific policy instruments for China's green industry (Y et al., 2021). By evaluating the scores of each policy tool, this research was able to gain insight into the trajectory and transformation of green industrial policy from both a temporal and spatial perspective. Based on the established research framework, this research selected policy rank, issuing authority, policy status, policy objectives, nature of policy, policy level, object of action, policy currency and scope of the policy as the core indicators to quantify China's green industrial policy (HANNIGAN et al., 2019).

In order to ensure the scientificity and rigor of the quantification process, this study compiles a guide to the quantification of green industrial policies by referring to the relevant policy literature and integrating the cutting-edge results in the field of policy quantification research. The guide lists the quantitative dimensions, criteria and their corresponding assignment methods in detail, which are summarized in the table below. It is worth noting that the scores of all quantitative indicators are directly derived from the policy text, which excludes the scoring bias caused by different participants. This not only eliminates the inconsistency of subjective scoring, but also ensures the accuracy of the policy composite score.

This study is based on the assumption that the quantitative dimensions of each policy instrument are of equal importance in the assessment and that they have a consistent impact on policy effectiveness (Zhengguang et al., 2023a, b; Chang and Andreoni, 2020). In view of this, this research used natural language processing (NLP) techniques to code the policies and assign values to the policy dimensions according to the "Guidelines for Quantifying Policy Instruments for the Green Industry". Overall, this Policy Effectiveness Scale provides a detailed and practical evaluation tool for policy research, and its careful categorization and quantitative criteria ensure an in-depth and comprehensive assessment of policies.

3.4. Coding of policies

Policy coding is the process of converting policy terms, objectives, or instruments into structured, quantifiable values or codes to facilitate subsequent data analysis and research (Lee and Kang, 2018). In this study, the coding format is "number - policy instrument - policy dimension". For articles containing multiple policy instruments, they were coded independently.

Faced with thousands of policy text data, this research introduced a high-precision policy coding strategy to ensure that each policy tool text is comprehensively and accurately captured, thus avoiding overlapping or missing data. First of all, the text preprocessing, including text cleaning, word segmentation, deletion of stop words, stemming extraction (restoration of word lines) and so on. Then key information and patterns are extracted from the text based on policy tool features. Third, policy coding, as it involves a large number of policy documents, this research used natural language processing (NLP) technology to automatically code and integrate the coding results into a unified dataset. For example, "0001-Tax Credit-Ministry of Science and Technology" or "0075-Energy Efficiency Increase-Local Regulations" or "0103-Increase Training-Modified," this coding ensures the rigor and objectivity of the policy analysis.

3.5. Quantification of policy effectiveness

This research constructed the PMC-TE index model, took the policy-coded dataset as the research object, combined with the quantification guide for green industry policy tools, and synthesized the multi-

Table 2
A Quantitative Guide to China's green industrial policy instruments.

Score	Detailed scoring standards	Score	Detailed scoring standards	Score	Detailed scoring standards
	(X1) Policy rank (Mingting et al., 2023; Adebayo and Ilesanmi, 2020)		(×4) Policy objectives (Nejat et al., 2015; Zhu et al., 2017)		(X7)Object of action (Liu et al., 2014; Acs et al., 2014)
5	Constitution and all laws	5	More than 10 times (including 10 times)	5	Long-term (more than 10 years)
4	Administrative regulations (including judicial interpretations)	4	5 to 10 (inclusive)	4	Medium to long term (5–10 years)
3	Local legislation	3	2 to 5 (inclusive)	3	Medium-term (3–5 years)
2	Departmental and local regulations	2	1 time	2	Short-term (1–3 years)
1	Other (including group regulations, industry regulations, etc.)	1	0th	1	1 year
	(X2) Issuing authority (Huang, 2010; Mergoni and De Witte, 2022)		(X5) Nature of policy (Flanagan et al., 2011; Yang and Huang, 2022)		(X8)Policy currency (Boqiang and Ting, 2023; Kuo et al., 2016)
5	The National People's Congress and its Standing Committee	5	Compulsory	5	Long-term (more than 10 years)
4	State Council (PRC)	4	Regulatory	4	Medium to long term (5–10 years)
3	Local people's congresses and their standing committees	3	Hybrid	3	Medium-term (3–5 years)
2	Ministries and local people's governments	2	Introductory	2	Short-term (1–3 years)
1	Else	1	Voluntary	1	1 year
	(X3) Policy status (Doernberg et al., 2019; Shen et al., 2021)		(X6) Policy level (Lillicrap et al., 2015)		(X9)Scope of the policy (Tey and Brindal, 2012; Mingting et al., 2023; Collins and Tabak, 2014)
5	Currently in force	5	National level	5	Over 15 industries
4	Not yet implemented	4	Provincial level	4	10-15 industries
3	Modified	3	Municipal	3	5-10 industries
2	Partially invalid	2	County level	2	3-5 industries
1	Lose effectiveness	1	Development zone level	1	1 industry

X1:The policy hierarchy reflects the legal status of the policy, from the Constitution and all laws to the hierarchy of group and industry regulations, ensuring that the authority and influence of the policy text is reasonably reflected.
X2:Publishing organization reflects the body that formulates and publishes the policy text, from the national level to the local and departmental level, which not only represents authority but also provides a glimpse of the scope of influence and level of implementation of the policy.
X3:Policy Status: this dimension emphasizes the real-time status of the policy, from currently effective to lapsed, and provides the researcher with a reference to the timeliness of the policy text.
×4:Policy objectives reflect the goal clarity and focus of the policy, and the number of times may represent the focus of the policy.
X5:Nature of the policy: The core features of the policy and the way it is implemented, ranging from mandatory to voluntary, are clarified, which helps to understand the implementation strategy and the expected effects of the policy.
X6:Policy level reveals the administrative level and scope of implementation of the policy, from the state to the development zone, showing us the depth of the policy's impact.
X7:The target of action emphasizes the coverage and audience of the policy, including government, enterprises, universities, research institutes, social groups, individuals, etc., which helps to understand the breadth and depth of the policy.
X8:Policy timeliness reveals the desired duration and life cycle of a policy, ranging from the long to the short term, providing us with the timeliness of the policy and possible strategic adjustments.
X9:The scope of the policy demonstrates the breadth and specificity of the policy in terms of its coverage of industries, ranging from multiple industries to a single industry.

dimensional, specific criteria and the corresponding assignment strategies to derive the detailed quantification results of different policy tools. Among them, the quantification guide for green industry policy tools mainly focuses on key indicators such as policy effectiveness rank, policy strength, policy effectiveness, policy objectives, policy tools and policy nature. Finally, this research analyzed in depth the development trajectory of each policy tool, the performance comparison of different types of policy tools, and their historical evolution logic.

The Policy Modeling Consistency (PMC) index model is based on the Omnia Mobilis assumption (Qin et al., 2023; Yuanfang, 2023). The assumption is that a wide range of complex variables in the model should be considered comprehensively, and the existence of each variable has a certain value, and the variables that have a small impact on the model should not be ignored. The advantages of the PMC index model in the evaluation of policies are (1) from the policy itself, enough to objectively reflect the strengths and weaknesses of the policy. (2) To analyze the internal heterogeneity and the level of strengths and weaknesses of a certain policy from a multi-dimensional perspective, to visually display the strengths and weaknesses of each dimension of the policy through the PMC surface diagram, and to analyze the specific meanings and levels of strengths and weaknesses of the variables, which is conducive to finding the weak points of the policy. (3) Combined with the theory of policy instruments, it is possible to assess a variety of problems in the evolution of policies.

However, in order to better capture the nuances and deep structure of policy texts, as well as to cope with the increasingly complex policy environment, this research proposed the PMC-TE (Policy Modeling Consistency-Text Encoder) index model. This model not only inherits the core idea and structure of the traditional PMC index model, but also introduces the "text encoder" technology, which enables it to more accurately analyze, encode, and quantify the key information and structure in policy texts.

The main innovation of the PMC-TE index model is its in-depth processing capability of text coding. By adopting advanced text coding technology, PMC-TE can not only capture the surface features of policy texts, but also dig deeper into their implied meanings and the logical relationships behind them, thus providing a more comprehensive and in-depth perspective for policy analysis and evaluation.

Therefore, this research quantitatively assessed China's green industrial policies through the PMC-TE index model. The analysis process is as follows: (1) construct the policy instrument pools based on the policy coding results; (2) assign values to the policy variables based on the quantitative guidelines of China's green industrial policy instruments, combined with the actual situation of the policy text; and (4) calculate the PMC-TE index for each policy instrument pool according to Equation (2), The PMC-TE index is calculated for each policy instrument pool. (X1 ∼ X9 are the 9 variable in Table 1.)

$$\mathrm{PMC-TE}=\begin{bmatrix} \mathrm{X1}\left(\sum_{i=1}^{5}\frac{\mathrm{X1i}}{5}\right)+\mathrm{X2}\left(\sum_{j=1}^{5}\frac{\mathrm{X2j}}{5}\right)+\mathrm{X3}\left(\sum_{k=1}^{5}\frac{\mathrm{X3k}}{5}\right)+ \\ \mathrm{X4}\left(\sum_{l=1}^{5}\frac{\mathrm{X4l}}{5}\right)+\mathrm{X5}\left(\sum_{m=1}^{5}\frac{\mathrm{X5m}}{5}\right)+\mathrm{X6}\left(\sum_{n=1}^{5}\frac{\mathrm{X5n}}{5}\right)+ \\ \mathrm{X7}\left(\sum_{p=1}^{5}\frac{\mathrm{X7p}}{5}\right)+\mathrm{X8}\left(\sum_{q=1}^{5}\frac{\mathrm{X8q}}{5}\right)+\mathrm{X9}\left(\sum_{r=1}^{5}\frac{\mathrm{X9r}}{5}\right) \end{bmatrix} \quad (2)$$

4. Results

4.1. Results for extraction of optimal policy topics

To ascertain the impact of parameter settings on the LDA model and to strengthen the credibility of our research findings, this study conducted a comprehensive sensitivity analysis. This analysis focused on varying the number of topics (k) and adjusting the hyperparameters α and β. The primary aim was to determine how these variations influence the model's ability to categorize text data effectively and coherently.

This study tested the LDA model with four different values for the number of topics:10, 20, 30, and 50. This range was selected to provide a broad overview of the model's performance across a spectrum of potential topic granularities. The hyperparameter β was consistently set at 0.01 across all models. The hyperparameter β was consistently set at 0.01 across all models. For α, this study employed a dynamic approach, setting it as 50/k, where k represents the number of topics. This approach allows α to adjust inversely with the number of topics, thereby enabling us to observe the effects of varying topic distribution within documents. This study raned the LDA model multiple times, each with a different combination of k, α, and β values as described above. For each model iteration, this study calculated the Perplexity and Coherence scores. The outcomes of these models were then compared to identify the settings that provided the most coherent and interpretable topics, while maintaining a reasonable level of predictive power.

Finally, as seen in Fig. 2-A, it demonstrates that the coherence score is optimized at 15 topics. Observing Fig. 2-B, there's a notable upward trend within the perplexity score in the range of 11–15 topics, and the perplexity score starts to plateau between 15 and 17 topics. By leveraging both perplexity and coherence as evaluation criteria, this research ascertained that the ideal number of topics for analyzing green industrial policies is 15.

The results (number of topics = 15) are deeply mined by unsupervised machine learning techniques to obtain a visual presentation of the LDA topic model of China's green industrial policy (shown in Fig. 3 and Fig. 4). In Figs. 3 and 4, each circle symbol represents a topic, and its size

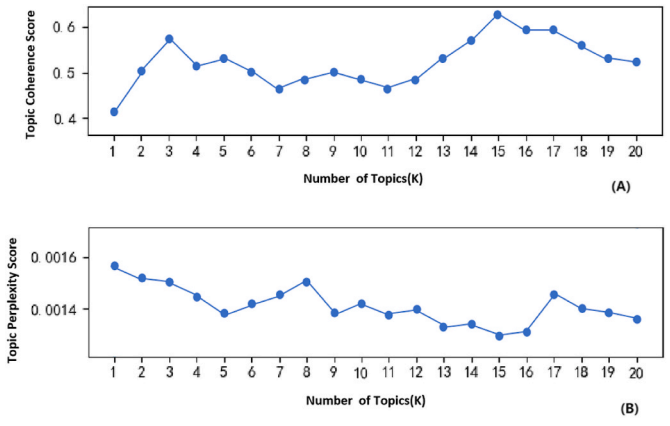


Fig. 2. LDA topic model output results (A):Coherence over number of topics result; (B): Perplexity over number of topics result.

reflects the frequency of occurrence of that topic, while the distance between the centers of the circles indicates the strength of association between two topics. Although the individual topic symbols overlap in the 2D visualization, this is only due to the visualization effect when mapping high-dimensional data to 2D space and does not imply actual topic overlap.

The visualization results of the topic model further demonstrate the frequency statistics of the feature words associated with the 15 topics in the policy text. Observing these topic feature words, this research finded that they are less homogeneous, which confirms the efficiency and accuracy of the segmentation, thus ensuring that the content reflected by the model is real and reliable. Fig. 3 presents the top 30 most frequent feature words in topic 1, and Fig. 4 shows the corresponding data in theme 15.

4.2. Results for identifying and clustering of policy instruments

4.2.1. Results for identifying policy instruments

Through in-depth cluster analysis of China's green industrial policy texts, this research identified 15 optimal themes. Each theme is closely associated with specific feature words that inform the description of the core characteristics of each policy tool. However, simply relying on the frequency of the top 30 words to determine the connotation of a policy tool is not sufficient. High-frequency words such as "green", "development", and "industry", despite their frequent occurrence in the policy texts, are not sufficient to accurately characterize the uniqueness of each policy tool. For this reason, this research integrated more high-frequency feature words, analyzed them in detail, and selected the most representative descriptions in order to clarify the categories of policy instruments corresponding to each theme. Based on these feature words and referring to Rothwdl and Zegveldt's theory of policy instruments, this research extracted 15 policy instruments for green industries, as shown in Table 3. These policy tools include technical support、information resource development、industrial parks and infrastructure development, among others.

4.2.2. Results for clustering of policy instruments

In addition, the distribution and clustering of themes, as illustrated in Fig. 5, provides us with further insight, suggesting that this research can reduce the dimension of the 15 policy instruments and cluster them twice. As a result of this process, this research categorized them into three main policy categories: themes 1, 2, 3, 6, 9, and 13 belong to the categoryI; themes 8, 11, 12, 14, and 15 are the categoryII; while themes 4, 5, 7, and 10 constitute the category III.

Based on the core content of the policy texts and policy instruments, this research further categorized the 15 policy instruments into three main categories: supply-based, demand-based and environmental, as shown in Table 4. Supply-side policies focus on the provision of resources, such as finance, talent and technology, and aim to stimulate the supply of green industries and enhance green products and services. Demand-based tools create market demand and encourage the adoption of green technologies, such as government procurement and demonstration programs. Environmental tools focus on environmental standards and regulation, such as target planning and regulations, to ensure that green production goes hand in hand with environmental protection. In short, supply-based tools promote green innovation, demand-based tools increase market demand, and environmental tools ensure the sustainability of green industries. The integrated application of these three is the core of China's green industrial policy.

China's green industrial policy is a complex policy system of strategies and methods used by the government to achieve strategic objectives and respond to public issues. These tools can be categorized according to their attributes and utility. Different policy instruments represent the interests and strategic orientations of policy makers. Policy-making requires diagnosing the problem, setting goals, and screening for appropriate tools. This process needs to take into account

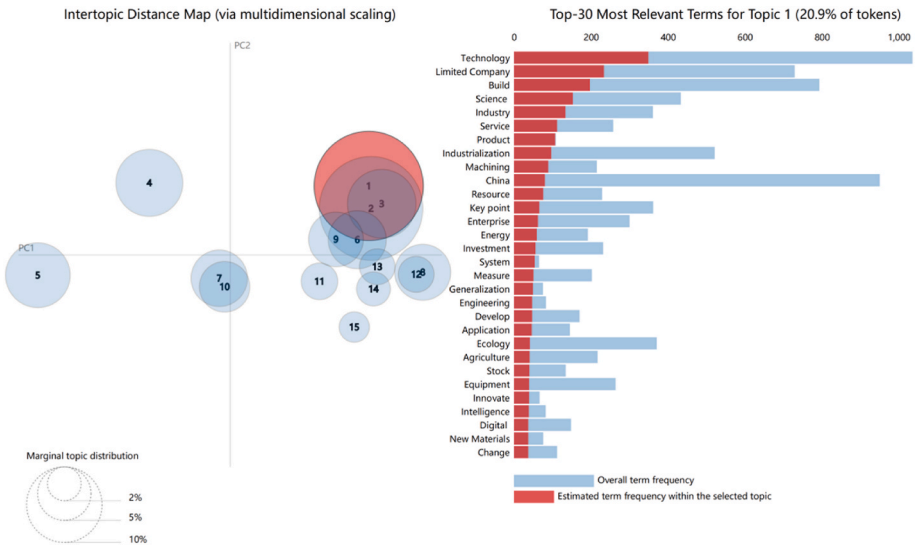


Fig. 3. Visualization results of LDA topic model for Topic 1.

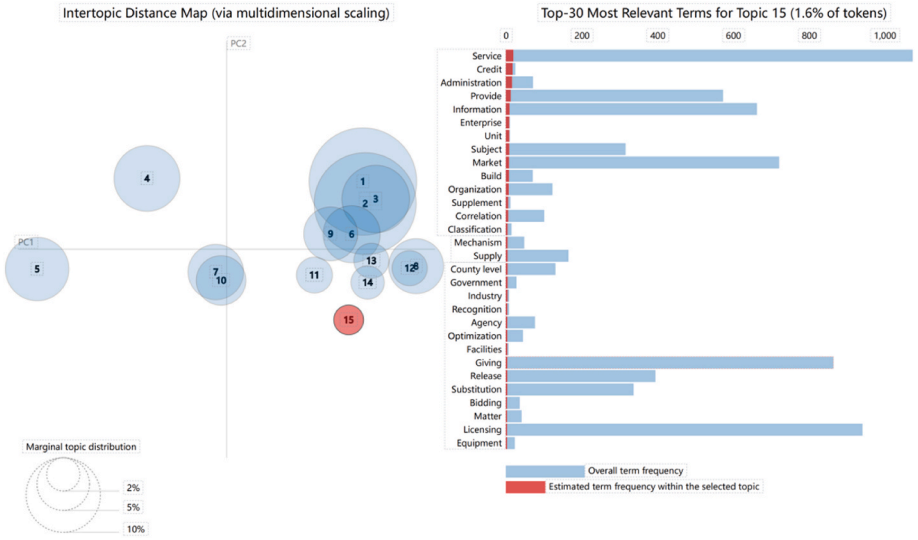


Fig. 4. Visualization results of LDA topic model for Topic 15.

government capacity, social repercussions, economic effects, and so on.

4.3. Results for qualification of different policy instruments

The policy process incorporates complex elements over time, and the PMC-TE Index emphasizes the need for policy measurement models to take into account a combination of variables, based on the principle that "everything is in motion and interrelated". The various policy instruments may exhibit significant variability in their design, implementation and management. Fig. 6 clearly shows that China's green industrial policy has developed and changed significantly from 1994 to 2022.

Overall, as shown in Fig. 6, China's green industrial policy has shown a clear upward trend, including an increase in the variety and number of policy instruments. This reflects the increasing importance that the Chinese Government attaches to the development of green industry, as well as the growing demand for diversified and refined policy instruments. This is in line with changes in China's economic, technological and social contexts, which require more nuanced policies to accommodate these changes.

Specifically, in Fig. 6, technology support policies came to prominence at an early stage, resource information building and demonstration project policies played an important role, and service outsourcing policies emerged later. Technical support policy was the earliest policy tool to emerge, and its notable effectiveness was also the greatest. In the primary stage of industrial development, China attached great importance to technological R&D and innovation, believing that technological innovation is the cornerstone of green industrial development. This also reflects China's attempts to steer the industry in a more efficient and environmentally friendly direction during that period. This was followed by the Resource Information Construction and Demonstration Project policy, which was identified by policymakers as a key means of accelerating the development of green industries, emphasizing the role of information sharing and the promotion of best practices in industrial development. The services outsourcing policy became effective around 2011, which is late compared to other policy instruments. On the one hand, by 2011, China had accumulated a certain amount of experience and talent in green technology R&D and green management, which provided a technological and human resource base for green service outsourcing. On the other hand, unlike the initial strategy of technology

Table 3
15 policy instruments and their definitions of China's green industrial policy.

ThemesNo.	Representative Subject Matter Words	Thematic Trait Words	Type of Policy Instrument	Definition of Policy Instruments
No.1	technical support; r&d support	technology; growth; development; development; core technology; development; innovation; science and technology; intelligent; digital technology; new materials.	Technical Support	Governments provide green industries with the support they need for technological innovation and upgrading.
No.2	information resource construction; industrial internet construction	internet of things; industrial internet; artificial intelligence; big data; resources; information; local; regional; positioning; scope	Information Resource Development	Provides information; data and knowledge required for the development of the green industry; helping companies better understand market dynamics; competitors and development opportunities.
No.3	industrial park restructuring; infrastructure optimization	housing; construction; resettlement; infrastructure; green industry; green fields; regions; development zones; industrial parks; industrial chains; supply chains; new businesses	Industrial Parks And Infrastructure Development	Provide a suitable development environment and infrastructure for green industries; and provide infrastructure support such as land; electricity and water supply to improve their productivity and reduce operating costs.
No.4	business incubation; transformation of scientific and technological achievements	small and medium-sized technology enterprises; scientific and technological achievements; transformation; industry-academia-research; precise docking; deep integration; transfer income ratio	Transformation Of Scientific And Technological Achievements	To support the transformation of scientific research results into practical environmental protection products and services; and to promote the commercialization of environmental protection science and technology.
No.5	property rights trading; intellectual property protection	intellectual property; investigation; analysis; inspection; examination; protection; property rights; reporting; complaints; rights defense	Intellectual Property Protection	Intellectual property rights (e.g.; patents; copyrights) protection for environmental protection technologies and innovations; and encouragement of R&D and innovation in environmental protection technologies by enterprises
No.6	financing; financial and tax support	credit; financial resources; funding; finance; authorization; recognition; assistance; sponsorship; investment tax; treasury; bonds	Financial Support	Governments provide direct or indirect financial support to green industries through fiscal policy instruments.
No.7	development planning; visioning	institutions; programs; planning; plans; regulations; burdens; mandates; objectives; visions; vision; governance; controls	Target Planning	A set of environmental and sustainability targets set up by the Government to steer businesses in a greener direction.
No.8	demonstration projects; demonstration bases; pilot areas	test area; site; ground; target; route; energy saving and environmental protection project; demonstration project; demonstration base; demonstration area	Demonstration Project	A category of government-funded or supported projects aimed at showcasing and promoting advanced green technologies and practices; including green buildings; renewable energy projects; and cleaner production demonstrations.
No.9	talent cultivation; innovation and entrepreneurship	training; entrepreneurship; job; work; labor; position; behavior; situation; position; function; occupation; security; culture	Talent Building	Developing; introducing and retaining personnel with specialized knowledge and skills related to green industry.
No.10	legal system; regulations; standards	laws; regulations; standards; provisions; norms; levels; systems; institutions; systems; responsibilities; measures; supervision; control	Regulatory And Standards Control	Setting environmental protection requirements through legislation and regulations; and compelling enterprises to comply with environmental laws; regulations and standards; including air quality standards; water quality standards; and waste disposal standards.
No.11	government procurement	replenish; replenish; recognize; agent; give; supply; supply; release; provide; substitute; license; equip	Government Procurement	The Government purchases green products and services from suppliers in the form of government procurement contracts to meet the needs of government agencies.
No.12	integrated foreign trade services; exchange control; foreign trade	pilot free trade zones; foreign trade; technology trade; foreign trade; foreign cultural trade; copyright trade; commerce; export.	Trade Control	It refers to the management and regulation of export; import and international trade policies; and governments can restrict or encourage international trade in green products through trade controls.
No.13	overseas warehouse; overseas operation center	offshore; offshore warehousing; cross border; e-commerce; groups; signatures; sections; offshore centers; overseas innovation institutions; overseas emergency assistance mechanisms	Overseas Organization Management	An institution established or supported by a government to manage international environmental protection and sustainable development affairs.
No.14	environmental labeling management; certification system; emission inspection	emissions; assessment; evaluation; functionality; energy efficiency; clean; new energy; inspection; qualification; green products	Environmental Labeling And Certification System Management	Used to assess and confirm that a product; service or business meets specific environmental standards and norms.
No.15	service outsourcing	assign; give; dispense; delegate; exercise; apply; execute; international contracting center; tender; protection	Service Outsourcing	The Government can encourage enterprises to outsource services in the field of environmental protection and sustainable development; such as environmental monitoring; waste treatment and energy-saving consultancy.

The theme numbers in this table correspond to the 15 themes in Fig. 3.

orientation and financial support, by 2011, the Chinese government believed that the development of green industries required a more market-oriented strategy, and service outsourcing could precisely provide more market opportunities for green enterprises, especially as developed countries began to put forward green requirements for external service providers while adopting green standards.

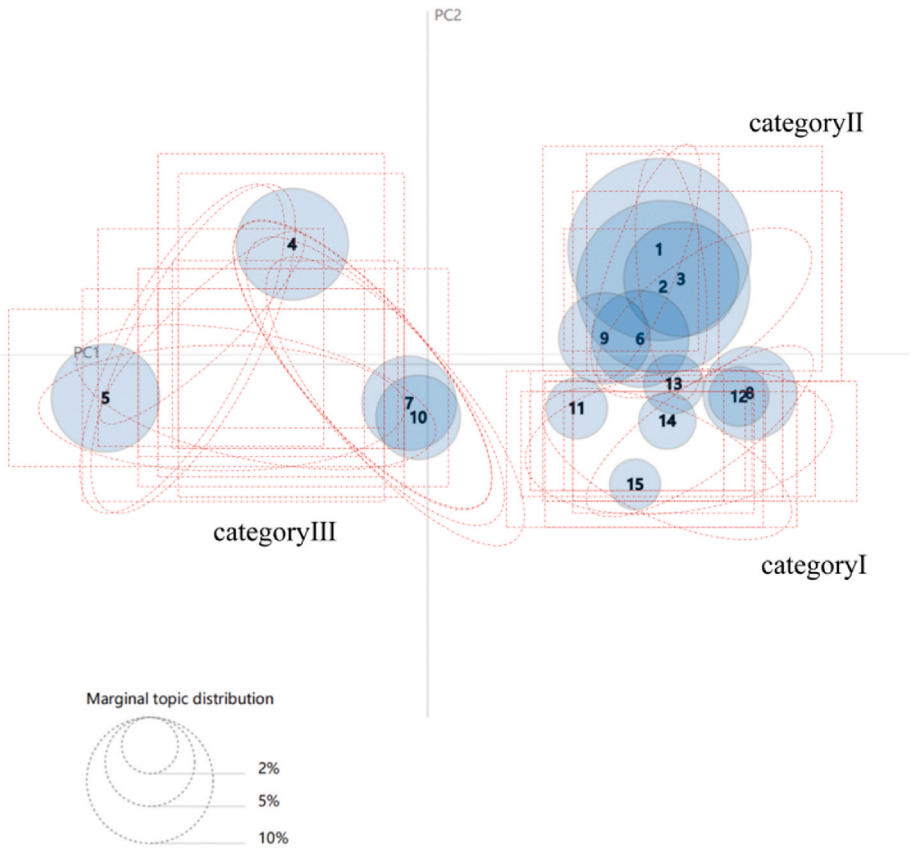


Fig. 5. Visualization results for clustering of China's green industrial policy instruments

Table 4
Supply–side, demand–side and environmental policy instruments about China's green industries.

Classification	Type of Policy Instrument
Supply-Side Policy	Financial Support
	Talent Building
	Technical Support
	Information Resource Development
	Industrial Parks And Infrastructure Development
Demand-Side Policy	Government Procurement
	Service Outsourcing
	Trade Control
	Environmental Labeling And Certification System
	Management
	Overseas Organization Management
	Demonstration Project
Environmental Policy	Target Planning
	Regulatory And Standards Control
	Intellectual Property Protection
	Transformation Of Scientific And Technological Achievements

4.4. Results for the quantification of different types of policies and their historical evolution

There may be significant differences in the formulation, implementation and management approaches of different policy types. In Fig. 7,supply-type policies occupy a dominant position, such as financial support, technical support, and talent development, which have become the dominant policies, reflecting the fact that the government's strategy in the green industry has always been biased towards providing the necessary resources and technological base to catalyze and promote the infrastructure and development of the green industry. Although demand-based policy instruments only began to appear in 2004, they grew rapidly between 2004 and 2016; however, demand-based policies such as government procurement and environmental labeling certification were relatively weak, indicating that market demand for green products and services had not yet been fully activated or promoted during this period, and that the government focused more on the construction of the supply side of the industry and the quality of the environment at this stage. Guarantee, and lacks guidance and stimulation of market demand. The simultaneous change in environment-based and supply-based policies after 2017 demonstrates the government's focus on maintaining and optimizing the ecological environment of the green industry, based on a response to ongoing economic and environmental sustainability challenges. It also suggests that as the industry matures, environmental considerations are seen as a strategic dimension as important as industry supply.

In addition, as shown in Figs. 6 and 7, from 1994 to 2022, China's green industrial policy exhibits obvious hierarchical and evolutionary features, which can be roughly categorized into four evolutionary stages. (1) Before 2002: the budding stage. Policies mainly focus on technical support and preliminary environmental protection measures. With the country's growing concern for environmental issues, the government gradually recognizes the importance of green industry in promoting sustainable development. (2) 2003–2015: Growth period. Especially with the 2008 Olympic Games and the subsequent series of international cooperation, green industry began to receive widespread attention. The government strengthened supply-oriented policies, promoted technological research and development, cultivated talents, and responded to initial market demand. Environmental regulations and standards have also gradually improved, providing clear guidelines for green industry development. (3) 2016–2019: Deepening phase. With the implementation of the "13th Five-Year Plan", green industry has been positioned as a strategic direction for national development. Supply-oriented policies have been further strengthened, while environment-

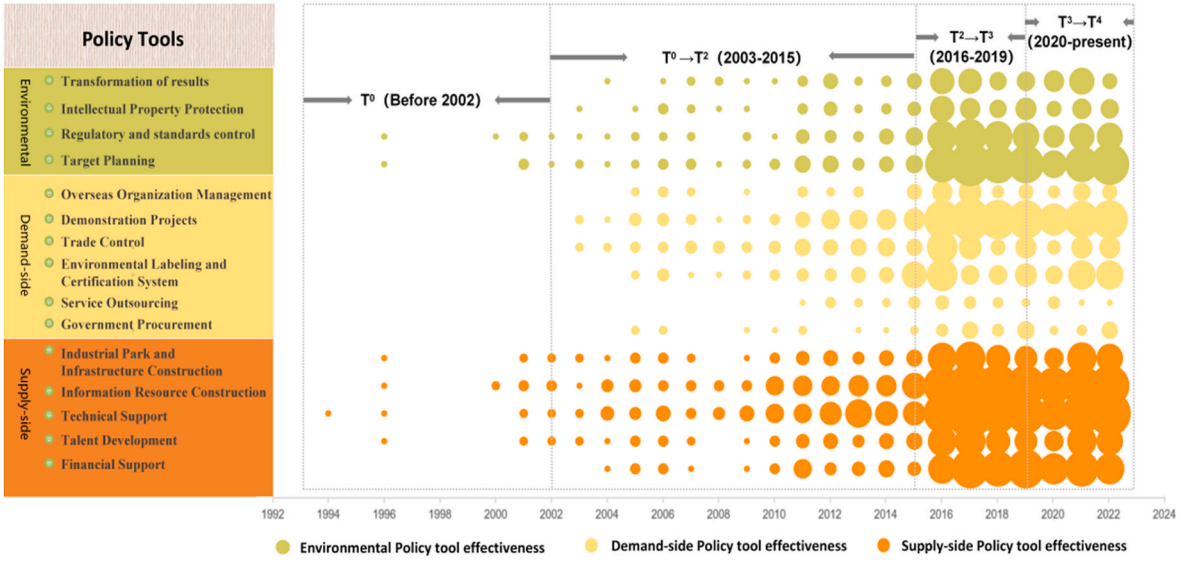


Fig. 6. Effectiveness of China's green industrial policy instruments

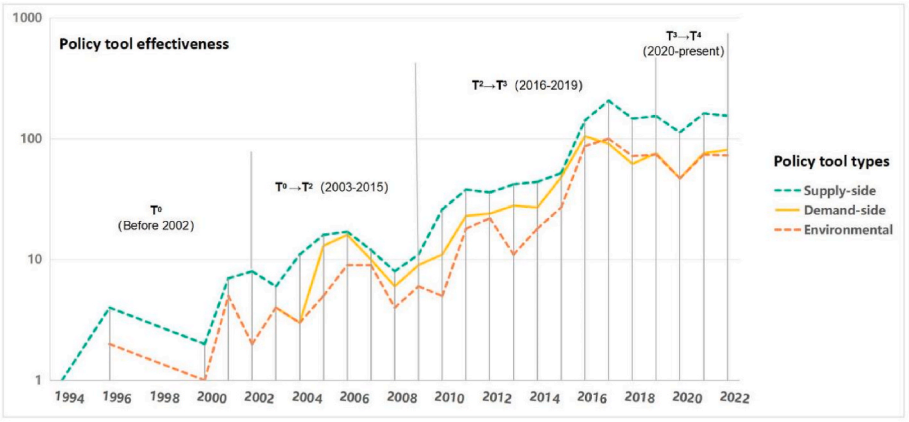


Fig. 7. Effectiveness of supply-, demand- and environment-oriented China's green industrial policy instruments

oriented policies such as target planning and regulation and standard control have begun to play a central role in ensuring high-quality industrial development and long-term environmental stability. (4) 2020 to the present: a period of refinement. Globally, there is an unprecedented rise in the importance attached to green and sustainable development, and China is no exception. Green industrial policies in this phase are more refined and systematic, not only further strengthening supply and environment-oriented policies, but also attempting to stimulate demand-oriented policies to promote green products and services in the market. In addition, as global cooperation to address climate change has grown, China's green industrial policy has been adjusted and optimized accordingly in this context.

The division of China's green industrial policy into four distinct stages is meticulously executed through a data-driven approach, anchored in three key analytical dimensions. Firstly, the synergy among demand-oriented, supply-oriented, and environment-oriented policies is scrutinized to identify shifts in policy focus and coherence. Secondly, the stages are informed by a detailed quantification of 15 policy instruments, considering both the volume and effectiveness of these tools over time. This analysis reveals patterns of concentration and potency that signify transitional phases. Lastly, significant temporal variations in policy types are closely examined, marking periods of notable evolution or change in policy direction. Collectively, these dimensions provide a comprehensive and dynamic understanding of the policy landscape, reflecting China's evolving commitment to sustainable industrial development and environmental stewardship. This multifaceted, data-driven methodology ensures a nuanced and accurate categorization of the stages in China's green industrial policy evolution.

5. Discussions

This study provides new insights into the evaluation and landscape of China's green industrial policies, revealing a dual narrative of commendable comprehensiveness intertwined with inherent systemic challenges. The analysis underscores the intricacies within the policy's hierarchical structure and its inter-policy dynamics, echoing the concerns raised by Xiang et al. (2023). The study identifies critical shortcomings, notably policy overlaps and gaps, which culminate in suboptimal guidance and disjointed implementation efforts (Allan et al., 2021). This is in stark contrast to the more streamlined and effective policy frameworks observed in countries like Germany, where a balanced approach between supply and demand side measures has led to more successful green industry outcomes (Pegels, 2017; Johnstone et al., 2021).

Our findings elucidate a marked emphasis within China's green industrial policies on supply-side mechanisms, predominantly technical support and financial incentives. This focus, while instrumental in fostering initial industrial capacity and technological advancement,

appears imbalanced when juxtaposed against the relatively nascent development of demand-side strategies. These demand-side components are indispensable in catalyzing market demand and fostering consumer participation in green initiatives. Their underdevelopment represents a critical gap, particularly when considering the holistic approach adopted by models such as the European Union's. Here, an integrated strategy harmonizing both supply and demand-side measures has been instrumental in cultivating dynamic markets for green technologies and sustainable products (Lucchese and Pianta, 2020; Pianta and Lucchese, 2020). This imbalance in policy focus might be symptomatic of a broader trend in green policy-making, where initial efforts often prioritize infrastructural and technical development. However, as the EU model illustrates, the maturation of green industries hinges on creating a balanced ecosystem that equally values market creation and consumer incentivization. The EU's approach underscores the necessity of fostering a culture of sustainability among consumers and businesses alike, an aspect that can significantly amplify the impact of supply-side advancements. By integrating lessons from such comprehensive policy frameworks, China can strategically recalibrate its focus, potentially spearheading a more holistic and effective trajectory in green industrial policy development.

Moreover, our analysis underscores systemic issues in the coherence and integration of China's green industrial policies across various developmental stages. While the policies are underpinned by well-defined strategic objectives, the fragmented execution often leads to suboptimal outcomes, as evidenced by the contrasting success of South Korea's cohesive and integrated green policy framework (Chen et al., 2021). This piecemeal approach not only undermines the efficacy of the policies but also instills a climate of uncertainty among businesses, impeding their ability to engage in long-term strategic planning and investment (Kang and Jo, 2021; Shah et al., 2020). The case of Japan offers instructive insights. Japan's systematic approach to policy management, characterized by regular assessments and iterative adjustments, has been pivotal in aligning its green industrial policies with contemporary challenges and opportunities (Ohta and Barrett, 2023; Tiberghien, 2023). The Japanese model demonstrates the effectiveness of a dynamic policy environment that is responsive to changing industrial landscapes and environmental imperatives. Emulating this model, China could benefit from instituting a mechanism for continual policy evaluation and refinement. This would not only enhance the internal consistency of the policies but also enable their evolution in tandem with technological advancements and market dynamics. Such a proactive and responsive policy framework could significantly bolster China's efforts in achieving its green industrial objectives, ensuring that the policies remain relevant, effective, and conducive to sustainable industrial growth.

The results of this study clearly indicate the necessity for a nuanced approach in formulating green industrial policies, one that accounts for the distinct attributes and requirements of diverse industrial sectors (Chen et al., 2023b). This tailored approach is crucial, as the efficacy of green policies is often contingent upon the specific nature of the industry they are designed to influence. Such specificity can address the unique challenges and opportunities within different sectors, ensuring that the policies are both relevant and effective. This perspective aligns with recent research underscoring the importance of sector-specific dynamics and organizational culture in determining policy outcomes (Imran et al., 2023; Saravade et al., 2023). Furthermore, our moderation analysis reveals a pivotal link between the level of environmental policy awareness in industries and the adoption of sustainable practices (Xu et al., 2023). Industries exhibiting greater awareness and understanding of environmental policies are more inclined to integrate sustainable practices into their operations. This trend mirrors observations in other industrialized nations, where increased policy awareness correlates with a higher uptake of eco-friendly practices (Hu and Zhang, 2023; Gao et al., 2023). These insights suggest that amplifying awareness and understanding of environmental policies could be a key strategy in fostering ecological sustainability across different industrial sectors. Cultivating a deeper comprehension of the implications and benefits of green policies could drive a more widespread and effective implementation of sustainable practices, significantly advancing environmental sustainability goals.

In conclusion, our study advocates for a sophisticated and adaptive strategy towards green industrial policy in China. The study's juxtaposition with international models serves as a mirror, reflecting how China can reshape its policy approach. By tackling present-day challenges and integrating exemplary practices from global frameworks, China stands to elevate the efficacy of its green industrial policies substantially. This approach not only fosters sustainable industrial growth but also reinforces environmental stewardship, thereby serving as a model for other countries navigating similar pathways towards sustainable development. Our findings underscore the potential of informed policy-making and effective policy assessment in shaping a greener future, resonating far beyond national borders.

6. Conclusions and future directions

6.1. Conclusions

After in-depth mining and comprehensive analysis of 7891 policy texts on China's green industry policies from 1994 to 2022, and adopting advanced research tools such as text mining technology, LDA topic model and PMC-TE index model this study provides a systematic categorization and quantitative assessment of China's green industrial policy instruments. The findings yield several critical insights:

(1) Our analysis identifies 15 pivotal policy instruments underpinning China's green industrial policy framework. Notably, technical support, information resource construction, and demonstration projects emerge as the most prevalent, underscoring their critical role in the policy landscape. These instruments not only manifest with the highest frequency across the analyzed texts but also exhibit profound impacts in terms of implementation effectiveness. This highlights their foundational importance in driving the agenda of green industrialization within the country. Conversely, while policies related to government procurement and service outsourcing have seen incremental growth in recent times, their relative influence and efficacy remain subdued in comparison. This discrepancy underscores the need for a more balanced policy toolset that enhances the synergistic effects of both supply and demand-side measures, ensuring a holistic approach to fostering a sustainable green industry.

(2) Based on the characteristics and objectives of these 15 policy instruments, this research have categorized them into three main groups: supply-type, demand-type and environment-type. Among them, supply-type policy instruments occupy a major position, followed by demand-type policy instruments, while environment-type policy instruments are relatively few. These three types of policy instruments complement and influence each other, with supply-based policies focusing on industrial infrastructure and technological innovation to provide "power" for the green industry, while demand-based policies stimulate the demand and expansion of the industry through market mechanisms and consumption orientation; and environmental policies ensure that the development of the green industry is consistent with the ecological environment and social responsibility. Environmental policies ensure that the development of green industries is consistent with ecological and social responsibility. The distribution of China's green industry policies across these three broad categories of policy instruments reflects a policy logic that extends from "industrial layout" to "market response" to "environmental governance".

(3) The research delineates four developmental phases of China's green industrial policy: inception, growth, consolidation, and maturation. From a timeline perspective, China's green industrial policy has gone through four key stages of development: the germination stage, the growth stage, the deepening stage, and the perfection stage, each of which corresponds to a different policy focus and implementation strategy. The germination phase lays the foundation for the subsequent phases and establishes the initial policy direction; the growth phase strengthens the implementation of the strategy and promotes the rapid expansion of the industry; the deepening phase focuses on refining the details of the policy and upgrading the quality of the industry; and the refinement phase marks a shift towards a systematic and sustained approach to the policy, which emphasizes the consistency of the green industry with the global sustainable development goals.

Theoretical Significance: This study theoretically broadens the methodological horizon of policy analysis by incorporating machine learning technologies such as the LDA model and PMC-TE model. This not only enhances the precision of policy analysis but also enriches the theoretical framework of policy research. Moreover, the traditional limitations of subjective interpretation in policy analysis are overcome through our approach, allowing for more accurate interpretation and utilization of complex and multilayered policy information.

Practical Significance: At the practical level, the quantitative analysis method of this study reduces the subjectivity inherent in expert scoring methods, increasing the transparency and objectivity of policy assessment, and rendering the evaluation of policy effects more scientific and credible. The classification and comparative analysis of policy tools provide policymakers with new insights into the strengths and limitations of various policy instruments, enabling the formulation of more effective policies. Furthermore, by identifying the most effective policy tools and practices, it ensures that resources are allocated to areas that have the most positive impact, optimizing policy outcomes.

In summary, this study advocates for a more refined and adaptive strategy towards China's green industrial policy. By addressing current challenges and integrating best practices from international examples, China can significantly enhance the efficacy of its green industrial policies. This approach not only promotes sustainable industrial development but also strengthens environmental protection, serving as a model for other countries on similar paths to sustainable development. Our findings emphasize the potential of informed policymaking and effective policy evaluation in shaping a greener future, with implications that extend well beyond national borders.

6.2. Future directions

This study offers an analysis based on the current dataset, yet green industry policies might evolve as new data emerges over time. While the research primarily centers on China's green industrial policy, it's crucial to recognize that the evolution of green industries is a global phenomenon. To extend the understanding, future studies might consider:

(1) Updating and incorporating newer datasets to refine policy impacts and effects.
(2) Delving deeper into the specific green industry policy instruments across China's diverse provinces, taking into account geographical variations and economic disparities for a more granular insight.
(3) Broadening the study's scope to include international comparisons, which would provide a holistic perspective on green industrial policy worldwide.

These potential directions can further enhance the depth and breadth of insights into the green industrial policy domain both theoretically and practically.

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CRediT authorship contribution statement

Chenchen Song: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Zhengguang Liu:** Software, Methodology. **Meng Yuan:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Congyu Zhao:** Visualization, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Acs, Z.J., Autio, E., Szerb, L., 2014. National systems of entrepreneurship: measurement issues and policy implications. Res. Pol. 43 (3), 476–494.
Adebayo, A.V., Ilesanmi, K.D., 2020. Financial resource management in the Nigerian public sector: policy measures to address loopholes. Probl. Perspect. Manag. 18 (2), 329.
Aikens, K., McKenzie, M., Vaughter, P., 2018. Environmental and Sustainability Education Policy Research: a Systematic Review of Methodological and Thematic Trends. Environmental and Sustainability Education Policy, pp. 265–292.
Allan, B., Lewis, J.I., Oatley, T., 2021. Green industrial policy and the global transformation of climate politics. Global Environ. Polit. 21 (4), 1–19.
Blei, D.M., Ng, A.Y., Jordan, M.I., 2003. Latent dirichlet allocation Journal of Machine Learning Research 3, 993–1022.
Bo, Zhu, Deng, Yuanyue, Hu, Xin, 2023. Global energy security: do internal and external risk spillovers matter? A multilayer network method, Energy Economics 126, 106961. https://doi.org/10.1016/j.eneco.2023.106961. ISSN 0140-9883.
Boqiang, Lin, Ting, Pan, 2023. Financing decision of heavy pollution enterprises under green credit policy: based on the perspective of signal transmission and supply chain transmission. J. Clean. Prod. 412, 137454. ISSN 0959-6526.
Cao, H., Wang, M., Su, S., et al., 2022. Explicit quantification of coastal cultural ecosystem services: a novel approach based on the content and sentimental analysis of social media. Ecol. Indicat. 137, 108756.
Chang, H.J., Andreoni, A., 2020. Industrial policy in the 21st century. Dev. Change 51 (2), 324–351.
Chen, Q., 2023. A review of textual analysis of policy tools for ecological space use control in China based on LDA thematic analysis model. Academic Journal of Management and Social Sciences 3 (1), 81–84.
Chen, L., Zhou, R., Chang, Y., Zhou, Y., 2021. Does green industrial policy promote the sustainable growth of polluting firms? Evidences from China. Sci. Total Environ. 764, 142927.
Chen, L., Hu, Z., Hu, X., et al., 2023a. How do green industrial policies accelerate regional sustainability transition? A spatiotemporal evaluation of policy with a relationalist perspective. J. Clean. Prod. 404, 136797.
Chen, M., Chen, R., Zheng, S., Li, B., 2023b. Green investment, technological progress, and green industrial development: implications for sustainable development. Sustainability 15 (4), 3808.
Collins, F.S., Tabak, L.A., 2014. Policy: nih plans to enhance reproducibility. Nature 505 (7485), 612–613.
Dan, Zhang, Zheng, Mingbo, Feng, Gen-Fu, Chang, Chun-Ping, 2022. Does an environmental policy bring to green innovation in renewable energy? Renew. Energy 195, 1113–1124. https://doi.org/10.1016/j.renene.2022.06.074. ISSN 0960-1481.
Dandan, Liu, Delu, Wang, 2022. Evaluation of the synergy degree of industrial decapacity policies based on text mining: a case study of China's coal industry. Resour. Pol. 76, 102547 https://doi.org/10.1016/j.resourpol.2021.102547. ISSN 0301-4207.
Das, S., Mondal, S., Puri, V., et al., 2022. Structural review of relics tourism by text mining and machine learning. Journal of Tourism, Heritage & Services Marketing (JTHSM) 8 (2), 25–34.

Di Giovanni, J., Rogers, J., 2023. The impact of US monetary policy on foreign firms. IMF Econ. Rev. 1–58.
Doernberg, A., Horn, P., Zasada, I., et al., 2019. Urban food policies in German city regions: an overview of key players and policy instruments. Food Pol. 89, 101782.
Dongyang, Zhang, Qunxi, Kong, 2022. Green energy transition and sustainable development of energy firms: an assessment of renewable energy policy. Energy Econ. 111, 106060. ISSN 0140-9883.
Duy, Nong, Heinz, Schandl, Yingying, Lu, 2023. George Verikios,Resource efficiency and climate change policies to support West Asia's move towards sustainability: a computable general equilibrium analysis of material flows. J. Clean. Prod. 421, 138458 https://doi.org/10.1016/j.jclepro.2023.138458. ISSN 0959-6526.
Fan, J., Ke, Y., Wang, K., 2020. Factor-adjusted regularized model selection. J. Econom. 216 (1), 71–85.
Flanagan, K., Uyarra, E., Laranja, M., 2011. Reconceptualizing the 'policy mix' for innovation. Res. Pol. 40 (5), 702–713.
Gao, W., Sebastiani, F., 2016. From classification to quantification in tweet sentiment analysis. Soc. Netw. Anal. Min. 6, 19. https://doi.org/10.1007/s13278-016-0327-z.
Gao, L., Tian, Q., Meng, F., 2023. The impact of green finance on industrial reasonability in China: empirical research based on the spatial panel Durbin model. Environ. Sci. Pollut. Control Ser. 30 (22), 61394–61410.
Ghasem, Layani, Saeed, Mehrjou, Zakariya, Farajzadeh, 2023. Effects of government policies reform on environmental sustainability: an integrated approach of PMP and system dynamics simulation model. J. Clean. Prod. 426, 138985. ISSN 0959-6526.
Ghosh, B., Papathanasiou, S., Dar, V., Kenourgios, D., 2022. Deconstruction of the green bubble during COVID-19 international evidence. Sustainability 14 (6), 3466. https://doi.org/10.3390/su14063466.
Giordano, T., 2015. Integrating industrial policies with innovative infrastructure plans to accelerate a sustainability transition. Environ. Innov. Soc. Transit. 14, 186–188.
Hamilton, M., 2017. Global, Regional and Local Infl Uences on Adult Literacy Policy in England//Adult and Lifelong Education. Routledge, pp. 122–138.
Hannigan, T.R., Haans, R., Vakili, K., et al., 2019. Topic modeling in management research: rendering new theory from textual data. Acad. Manag. Ann. 13 (2), 586–632.
Harrison, A., Martin, L.A., Nataraj, S., 2017. Green industrial policy in emerging markets. Annual Review of Resource Economics 9, 253–274.
Hayat, A., Sohail, M., El Jery, A., Al-Zaydi, K.M., Raza, S., Ali, H., et al., 2023a. Recent advances in ground-breaking conjugated microporous polymers-based materials, their synthesis, modification and potential applications. Mater. Today.
Hayat, A., Sohail, M., El Jery, A., Al-Zaydi, K.M., Raza, S., Ali, H., et al., 2023b. Recent Advances, Properties, Fabrication and Opportunities in Two-Dimensional Materials for Their Potential Sustainable Applications. Energy Storage Materials, 102780.
Hayat, A., Sohail, M., Moussa, S.B., Al-Muhanna, M.K., Iqbal, W., Ajmal, Z., et al., 2023c. State, synthesis, perspective applications, and challenges of Graphdiyne and its analogues: a review of recent research. Adv. Colloid Interface Sci. 319, 102969.
Hu, J., Zhang, H., 2023. Has green finance optimized the industrial structure in China? Environ. Sci. Pollut. Control Ser. 30 (12), 32926–32941.
Hu, Y.J., Zhang, R., Wang, H., Li, C., Tang, B.J., 2024. Synergizing policies for carbon reduction, energy transition and pollution control: evidence from Chinese power generation industry. J. Clean. Prod., 140460
Huang, Honghua, 2010. The emergence of policy instrument theory and its development in China. Soc. Sci. (4), 13–19+187.
Huaping, Sun, Bless Kofi, Edziah, Chuanwang, Sun, Anthony Kwaku, Kporsu, 2019. Institutional quality, green innovation and energy efficiency. Energy Pol. 135, 111002 https://doi.org/10.1016/j.enpol.2019.111002. ISSN 0301-4215.
Imran, M., Hayat, N., Saeed, M.A., Sattar, A., Wahab, S., 2023. Spatial green growth in China: exploring the positive role of investment in the treatment of industrial pollution. Environ. Sci. Pollut. Control Ser. 30 (4), 10272–10285.
Jelodar, H., Wang, Y., Yuan, C., et al., 2019. Latent Dirichlet allocation (LDA) and topic modeling: models, applications, a survey. Multimed. Tool. Appl. 78, 15169–15211.
Johnstone, P., Rogge, K.S., Kivimaa, P., Fratini, C.F., Primmer, E., 2021. Exploring the re-emergence of industrial policy: perceptions regarding low-carbon energy transitions in Germany, the United Kingdom and Denmark. Energy Res. Social Sci. 74, 101889.
Kang, N., Jo, K., 2021. State–business relations in flux: capturing the structural power of business in South Korea's green industrial policy. J. Contemp. Asia 51 (5), 713–736.
Karaman, C., Karaman, O., Show, P.L., Orooji, Y., Karimi-Maleh, H., 2022. Utilization of a double-cross-linked amino-functionalized three-dimensional graphene networks as a monolithic adsorbent for methyl orange removal: equilibrium, kinetics, thermodynamics and artificial neural network modeling. Environ. Res. 207, 112156.
Karimi-Maleh, H., Liu, Y., Li, Z., Darabi, R., Orooji, Y., Karaman, C., et al., 2023a. Calf thymus ds-DNA intercalation with pendimethalin herbicide at the surface of ZIF-8/Co/rGO/C3N4/ds-DNA/SPCE; A bio-sensing approach for pendimethalin quantification confirmed by molecular docking study. Chemosphere 332, 138815.
Karimi-Maleh, H., Orooji, Y., Karimi, F., Karaman, C., Vasseghian, Y., Dragoi, E.N., Karaman, O., 2023b. Integrated approaches for waste to biohydrogen using nanobiomediated towards low carbon bioeconomy. Adv. Compos. Hybrid Mater. 6 (1), 29.
Kiss, T., Popovics, S., 2021. Evaluation on the effectiveness of energy policies–Evidence from the carbon reductions in 25 countries. Renew. Sustain. Energy Rev. 149, 111348.
Kuo, C.F.J., Lin, C.H., Hsu, M.W., 2016. Analysis of intelligent green building policy and developing status in Taiwan. Energy Pol. 95, 291–303.
Kyle, S., 2023. Herman Green growth and innovation in the Global South: a systematic literature review. Innovation and Development 13 (1), 43–69. https://doi.org/10.1080/2157930x.2021.1909821.
Lee, H., Kang, P., 2018. Identifying core topics in technology and innovation management studies: a topic model approach. J. Technol. Tran. 43 (5), 1291–1317.
Lillicrap, T.P., Hunt, J.J., Pritzel, A., et al., 2015. Continuous control with deep reinforcement learning. arXiv preprint arXiv, 1509.02971.
Liu, F., Liu, Z., 2022. Quantitative evaluation of waste separation management policies in the Yangtze River Delta Based on the PMC index model. Int. J. Environ. Res. Publ. Health 19 (7), 3815.
Liu, Y., Fang, F., Li, Y., 2014. Key issues of land use in China and implications for policy making. Land Use Pol. 40, 6–12.
Liu, Z., Wang, Z.L., Yuan, C.J., 2022. Impact of independent technological innovation on industrial carbon emissions and trend prediction from the perspective of structure. China Population, Resources and Environment 32 (7), 12–21.
Lucchese, M., Pianta, M., 2020. Europe's Alternative: a Green Industrial Policy for Sustainability and Convergence.
Mandy, Meng Fang, 2021. Old wine in a new bottle? Green industrial policy and the use of safeguards in the solar sector. J. World Trade 55 (4), 573–596.
Matheus, Koengkan, José Alberto, Fuinhas, Emad, Kazemzadeh, Nooshin, Karimi Alavijeh, Saulo Jardim, de Araujo, 2022. The impact of renewable energy policies on deaths from outdoor and indoor air pollution: empirical evidence from Latin American and Caribbean countries. Energy 245 (123209). https://doi.org/10.1016/j.energy.2022.123209. ISSN 0360-5442.
Matthias van den, Heuvel, David, Popp, 2023. The role of venture capital and governments in clean energy: lessons from the first cleantech bubble. Energy Econ. 124, 106877 https://doi.org/10.1016/j.eneco.2023.106877. ISSN 0140-9883.
Mergoni, A., De Witte, K., 2022. Policy evaluation and efficiency: a systematic literature review. Int. Trans. Oper. Res. 29 (3), 1337–1359.
Ming, Gao, 2023. The impacts of carbon trading policy on China's low-carbon economy based on county-level perspectives. Energy Pol. 175, 113494 https://doi.org/10.1016/j.enpol.2023.113494. ISSN 0301-4215.
Mingting, Kou, Yang, Yuanqi, Feng, Ze, Chen, Kaihua, Wei, Jianwu, 2023. Are incentives excessive or insufficient? The impact of R&D fiscal policies on R&D inefficiency in China. Technol. Forecast. Soc. Change 194, 122707. https://doi.org/10.1016/j.techfore.2023.122707. ISSN 0040-1625.
Mølstad, C.E., Karseth, B., 2016. National curricula in Norway and Finland: the role of learning outcomes. Eur. Educ. Res. J. 15 (3), 329–344. https://doi.org/10.1177/1474904116639311.
Naqvi, B., Rizvi, S.K.A., Mirza, N., et al., 2023. Financial market development: a potentiating policy choice for the green transition in G7 economies. Int. Rev. Financ. Anal. 87, 102577.
Nejat, P., Jomehzadeh, F., Taheri, M.M., et al., 2015. A global review of energy consumption, CO2 emissions and policy in the residential sector (with an overview of the top ten CO2 emitting countries). Renewable and sustainable energy reviews 43, 843–862.
Ohta, H., Barrett, B., 2023. Politics of climate change and energy policy in Japan: is green transformation likely? Earth System Governance 17, 100187.
Pegels, A., 2017. Germany: the energy transition as a green industrial development agenda. Green Industrial Policy.
Philip, McCann, Ortega-Argilés, Raquel, Sevinc, Deniz, Cepeda-Zorrilla, Magda, 2023. Rebalancing UK regional and industrial policy post-Brexit and post-Covid-19: lessons learned and priorities for the future. Reg. Stud. 57 (6), 1113–1125. https://doi.org/10.1080/00343404.2021.1922663.
Pianta, M., Lucchese, M., 2020. Rethinking the European Green Deal: an industrial policy for a just transition in Europe. Rev. Radic. Polit. Econ. 52 (4), 633–641.
Qin, Liu, Jia, Mengting, De Xia, 2023. Dynamic evaluation of new energy vehicle policy based on text mining of PMC knowledge framework. J. Clean. Prod. 392, 136237 https://doi.org/10.1016/j.jclepro.2023.136237. ISSN 0959-6526.
Qiu Yue, Zhang, Best, Rohan, Chareunsy, Andrea, 2023. The impact of globalisation and education in promoting policies for renewables and energy efficiency. J. Clean. Prod. 421, 138559. ISSN 0959-6526.
Ruiz Estrada, M.A., Yap, S.F., Nagaraj, S., 2007. Beyond the Ceteris Paribus Assumption: Modeling Demand and Supply Assuming Omnia Mobilis.
Saravade, V., Chen, X., Weber, O., Song, X., 2023. Impact of regulatory policies on green bond issuances in China: policy lessons from a top-down approach. Clim. Pol. 23 (1), 96–107.
Shah, I.H., Dong, L., Park, H.S., 2020. Tracking urban sustainability transition: an eco-efficiency analysis on eco-industrial development in Ulsan, Korea. J. Clean. Prod. 262, 121286.
Shen, Wei, He, Jijiang, Yao, Shangheng, 2021. Green industrial policy in the post grid parity era: governing integrated Solar+ projects in China. Energy Pol. 150, 112129 https://doi.org/10.1016/j.enpol.2020.112129. ISSN 0301-4215.
Steffen, B., Patt, A., 2022. A historical turning point? Early evidence on how the Russia-Ukraine war changes public support for clean energy policies. Energy Res. Social Sci. 91, 102758.
Tang, K., Qiu, Y., Zhou, D., 2020. Does command-and-control regulation promote green innovation performance? Evidence from China's industrial enterprisesSci. Total Environ 712. Article 136362, 0.1016/j.scitotenv.2019.136362.
Tey, Y.S., Brindal, M., 2012. Factors influencing the adoption of precision agricultural technologies: a review for policy implications. Precis. Agric. 13, 713–730.
Tiberghien, Y., 2023. Kishida's climate policy and opportunities for US-Japan cooperation. Asia Policy 30 (1), 49–65.
Tichenor, M., Merry, S.E., Grek, S., et al., 2022. Global public policy in a quantified world: sustainable Development Goals as epistemic infrastructures. Policy and Society 41 (4), 431–444.
Usmani, R.A., Mohammad, A.S., Ansari, S.S., 2023. Comprehensive biofuel policy analysis framework: a novel approach evaluating the policy influences. Renew. Sustain. Energy Rev. 183, 113403.
Victor, H., 2023. Cantú, Antonin Ponsich, Catherine Azzaro-Pantel, Eduardo Carrera, Capturing spatial, time-wise and technological detail in hydrogen supply chains: a

bi-level multi-objective optimization approach. Appl. Energy 344, 121159. https://doi.org/10.1016/j.apenergy.2023.121159. ISSN 0306-2619.
Wang, Y., Zhang, W., 2023. Green credit policy, market concentration and green innovation: empirical evidence from local governments' regulatory practice in China. J. Clean. Prod., 140228
Wu, S., Wu, L., Zhao, X., 2022. Impact of the green credit policy on external financing, economic growth and energy consumption of the manufacturing industry. Chinese Journal of Population, Resources and Environment 20 (1), 59–68.
Xiang, Z., Hasan, A., Mao, D., Saeed, M.K., Mirza, F., 2023. Unraveling the impact of eco-centric leadership and pro-environment behaviors in healthcare organizations: role of green consciousness. J. Clean. Prod., 139704
Xin, Cui, Wang, Chunfeng, Ahmet, Sensoy, Liao, Jing, Xie, Xiaochen, 2023. Economic policy uncertainty and green innovation: evidence from China. Econ. Modell. 118, 106104 https://doi.org/10.1016/j.econmod.2022.106104. ISSN 0264-9993.
Xu, L., Yang, L., Li, D., Shao, S., 2023. Asymmetric effects of heterogeneous environmental standards on green technology innovation: evidence from China. Energy Econ. 117, 106479.
Y, Li, Chen, Y., Wang, Q., 2021. Evolution and diffusion of information literacy topics Scientometrics 126 (5), 4195–4224.
Yang, Chao, Huang, Cui, 2022. Quantitative mapping of the evolution of AI policy distribution, targets and focuses over three decades in China. Technol. Forecast. Soc. Change 174, 121188. https://doi.org/10.1016/j.techfore.2021.121188. ISSN 0040-1625.
Yang, C., Bu, S., Fan, Y., Wan, W.X., Wang, R., Foley, A., 2023. Data-driven prediction and evaluation on future impact of energy transition policies in smart regions. Appl. Energy 332, 120523.
Yeo, Y., Oh, I., 2023. Evaluating the impacts of renewable energy promotion policies on sustainable development: a computable general equilibrium model approach. J. Clean. Prod. 421, 138360.
Yuan, M., Thellufsen, J.Z., Lund, H., Liang, Y., 2020a. The first feasible step towards clean heating transition in urban agglomeration: a case study of Beijing-Tianjin-Hebei region. Energy Convers. Manag. 223, 113282.
Yuan, M., Zhang, H., Wang, B., Huang, L., Fang, K., Liang, Y., 2020b. Downstream oil supply security in China: policy implications from quantifying the impact of oil import disruption. Energy Pol. 136, 111077.
Yuan, M., Zhang, H., Wang, B., Zhang, Y., Zhou, X., Liang, Y., 2020c. Future scenario of China's downstream oil reform: improving the energy-environmental efficiency of the pipeline networks through interconnectivity. Energy Pol. 140, 111403.
Yuan, M., Thellufsen, J.Z., Lund, H., Liang, Y., 2021. The electrification of transportation in energy transition. Energy 236, 121564.
Yuan, M., Sorknæs, P., Lund, H., Liang, Y., 2022. The bidding strategies of large-scale battery storage in 100% renewable smart energy systems. Appl. Energy 326, 119960.
Yuanfang, Z., 2023. Optimization of financial market risk prediction system based on computer data simulation and Markov chain Monte Carlo. Soft Comput. 1–10.
Zahedi, Rahim, Zahedi, Alireza, Abolfazl Ahmadi, 2022. Strategic study for renewable energy policy, optimizations and sustainability in Iran. Sustainability 14 (4), 2418. https://doi.org/10.3390/su14042418.
Zhaohui, Zhu, Yafei, Tan, 2022. Can green industrial policy promote green innovation in heavily polluting enterprises? Evidence from China. Econ. Anal. Pol. 74, 59–75. https://doi.org/10.1016/j.eap.2022.01.012. ISSN 0313-5926.
Zhaotian, Chong, Qunwei, Wang, Lei, Wang, 2023. Is the photovoltaic power generation policy effective in China? A quantitative analysis of policy synergy based on text mining. Technol. Forecast. Soc. Change 195, 122770. https://doi.org/10.1016/j.techfore.2023.122770. ISSN 0040-1625.
Zhengguang, Liu, Zhiling, Guo, Song, Chenchen, Du, Ying, Chen, Qi, Chen, Yuntian, Zhang, Haoran, 2023a. Business model comparison of slum-based PV to realize low-cost and flexible power generation in city-level. Appl. Energy 344, 121220. ISSN 0306-2619.
Zhengguang, Liu, Guo, Zhiling, Chen, Qi, Song, Chenchen, Shang, Wenlong, Meng, Yuan, Zhang, Haoran, 2023b. A review of data-driven smart building-integrated photovoltaic systems: challenges and objectives. Energy 263 (Part E), 126082. ISSN 0360-5442.
Zhu, Y., Mottaghi, R., Kolve, E., et al., 2017. Target-driven visual navigation in indoor scenes using deep reinforcement learning[C]//2017 IEEE international conference on robotics and automation (ICRA). ieee 3357–3364.